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Higher-Order Root Finding as a Hierarchy of Scales

Newton, Halley, and the Householder family read off the characteristic length scales of a function

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Newton's method steps by $-f/f'$. Read f/f' as a length, not as a ratio of derivatives: it is the characteristic scale L_f of the exponential that matches f at the current point, so Newton steps one scale-length toward the root. Every higher-order method is that same step, corrected for how the scale drifts as you move. We make the correction explicit and read off three pictures of it. Define a tower of scales $L_k = f^{(k)}/f^{(k+1)}$ and the ratios $\rho_k = L_{k-1}/L_k$ between rungs. Halley is Newton divided by $1 - \frac{1}{2}\rho_1$; the order-d Householder method is L_f times a fixed rational function of $\rho_1, \ldots, \rho_{d-1}$ — one continued fraction, read one rung deeper at each order. The step is a clock, the time-to-contact of a particle closing on a target; and because a root of f is a pole of $1/f$, it is the ratio test ranging to that pole. We close with a root finder that needs only the recursion for the coefficients of $1/f$ and one ratio per step.

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Newton's method is a statement about length

Newton's method updates a guess x_t for a root of f by

$$x_{t+1} = x_t - \frac{f(x_t)}{f'(x_t)} = x_t - \frac{1}{f'(x_t)/f(x_t)}. \tag{1}$$

The textbook reading is algebraic: solve the local linear model $f(x) + f'(x)(y - x) = 0$ for its zero. Here is a different reading, the one that generalizes. The ratio f'/f is the logarithmic derivative, and it has units of inverse length. So its reciprocal,

$$L_f(x) \;\coloneqq\; \frac{f(x)}{f'(x)}, \tag{2}$$

is a length — and it is *the* length built into f at x. Fit a local exponential $f(y) \approx A\,e^{y/L}$ to the value and slope of f at x, and the fit hands back $L = L_f(x)$, because $g/g' = L$ everywhere for $g = A\,e^{y/L}$. So L_f is the distance over which f changes by a factor of e, and Newton's rule (1) reads:

> *Step one characteristic length toward the root:* $x_{t+1} = x_t - L_f(x_t)$.

That is more than a mnemonic. The exponential is the one function whose scale never changes, and it is exactly the function on which a single scale-length is the right unit of progress. Real functions have a scale that drifts as x moves, and that drift is what Newton's first-order step gets wrong. Every higher-order method is bookkeeping for the drift.

Here is the plan. Section 2 makes the drift quantitative. Section 3 shows Halley is Newton with the first drift correction. Section 4 reads the step as a clock — the physical heart of the paper. Section 5 writes the whole Householder family as one continued fraction in the scale ratios. Section 6 turns the integer order into a continuous dial and finds the root sitting where $1/f$ blows up. Section 7 folds all of it into an algorithm you can run. Don't worry if the length picture feels loose now; each section sharpens it.

The scale drifts, and that drift is what Newton misses

If $L_f = f/f'$ is the scale of f, then f' has its own scale f'/f'', and f'' has f''/f''', on up. Define the tower

$$L_k(x) \;\coloneqq\; \frac{f^{(k)}(x)}{f^{(k+1)}(x)}, \qquad k = 0, 1, 2, \ldots, \tag{3}$$

so $L_0 = L_f$ is the leading scale and each L_k is the characteristic length of the k-th derivative. The same information as the derivatives $f, f', f'', \ldots$, carried in the currency of lengths.

A single length can't tell you how the scale is changing; for that you need the dimensionless ratio between neighboring rungs,

$$\rho_k(x) \;\coloneqq\; \frac{L_{k-1}(x)}{L_k(x)}, \qquad k = 1, 2, \ldots. \tag{4}$$

The first one is worth seeing in full:

$$\rho_1 \;=\; \frac{L_0}{L_1} \;=\; \frac{f/f'}{f'/f''} \;=\; \frac{f\,f''}{(f')^2}. \tag{5}$$

Two reference functions fix its meaning. For a *line* $f = a(x - x^\star)$ we have $f'' = 0$, so $\rho_1 = 0$: the scale isn't bending. For an *exponential* $f = e^{x/L}$ every rung equals L and every $\rho_k = 1$: the scale is perfectly self-similar. In between, ρ_1 measures how far f has bent from a straight line toward an exponential.

Now the one quantity Newton ignores. Differentiate the leading scale (2):

$$L_f'(x) = \frac{(f')^2 - ff''}{(f')^2} = 1 - \frac{ff''}{(f')^2} = 1 - \rho_1. \tag{6}$$

The leading scale drifts at rate $L_f' = 1 - \rho_1$. In plain terms: Newton steps by $-L_f$ as if L_f were the fixed distance to the root, but L_f moves as x moves, and ρ_1 says how fast. A line has $L_f' = 1$ — the scale equals the distance to the root and runs out right at it. An exponential has $L_f' = 0$ — the scale never moves. So the first correction any second-order method makes can depend on nothing but ρ_1. That points straight at Halley.

Halley repairs the drift

Halley's second-order method, in these variables, is Newton with one correction factor:

$$x_{t+1} = x_t - \frac{L_f}{1 - \frac{1}{2}\,\dfrac{L_f}{L_1}} = x_t - \frac{L_f}{1 - \frac{1}{2}\,\rho_1}. \tag{7}$$

This is the standard Halley iteration [4, 5]. To see it, take the textbook form and divide top and bottom by $2(f')^2$:

$$x_{t+1} = x_t - \frac{2ff'}{2(f')^2 - ff''} = x_t - \frac{f/f'}{1 - \dfrac{ff''}{2(f')^2}} = x_t - \frac{L_f}{1 - \frac{1}{2}\rho_1}, \tag{8}$$

using (5) for the last step. Halley keeps Newton's direction — the leading scale L_f — and rescales the step by $1/(1 - \frac{1}{2}\rho_1)$, a factor built entirely from the first scale ratio. When $\rho_1 > 0$ (the scale shrinking ahead of the step) it lengthens Newton's step; when $\rho_1 < 0$ it shortens it.

Halley is Newton plus a first-order correction for the drift. It accounts for how the length scale changes under the very step it is about to take. Before climbing to higher orders, look at what this step *is*, physically.

The kinematic reading: a clock counting down to contact

Why does $-f/f'$ keep showing up? Because it is a time.

Picture f along x as the signed miss-distance of a particle closing on a target, with each step of x playing the role of elapsed time. Then the local derivatives are the particle's state of motion: f is position, f' is velocity — the closing rate — f'' is acceleration, f''' is jerk, and on up the tower.

In those terms the characteristic length is a clock. Distance over closing rate is the first-order time to contact,

$$\tau = -\frac{f}{f'} = -L_f, \tag{9}$$

the same $-L_f$ Newton steps by — and the same quantity that visual guidance calls "τ" and missile guidance calls time-to-go [7].

Newton steps by the time-to-contact at constant closing speed.

The drift law says this clock keeps honest time. Since $L'_f = 1 - \rho_1$, the time-to-contact changes with position at rate $\mathrm{d}\tau/\mathrm{d}x = \rho_1 - 1$, which is exactly -1 when the motion is uniform ($\rho_1 = 0$): advance one step, lose one tick. The dimensionless acceleration $\rho_1 = ff''/(f')^2$ is the correction when the closing rate is not constant, and Halley (7) folds it in: $\tau = \tau_1/(1 - \frac{1}{2}\rho_1)$ is the time-to-contact corrected for acceleration. Each higher order adds the next term of the motion — order 3 the jerk, order 4 the snap — so the order-d method reads the contact time off the kinematics through the d-th derivative. The next section gives that family a single form.

The whole family is one continued fraction

Newton and Halley are the first two rungs of one family. The order-d Householder method is

$$x_{t+1} = x_t + d\,\frac{\big(1/f\big)^{(d-1)}}{\big(1/f\big)^{(d)}}, \tag{10}$$

with $d = 1$ Newton and $d = 2$ Halley. Each member is a ratio of consecutive derivatives of $1/f$, and dividing through by the leading scale collapses all of that into the dimensionless ratios: the order-d step is $-L_f$ times a fixed rational function of $\rho_1, \ldots, \rho_{d-1}$. The first three:

$$\text{Newton } (d=1): \quad x_{t+1} = x_t - L_f, \tag{11}$$

$$\text{Halley } (d=2): \quad x_{t+1} = x_t - L_f\,\frac{1}{1 - \frac{1}{2}\rho_1}, \tag{12}$$

$$\text{order } 3: \quad x_{t+1} = x_t - L_f\,\frac{1 - \frac{1}{2}\rho_1}{1 - \rho_1 + \frac{1}{6}\rho_1^2\rho_2}. \tag{13}$$

The order-3 denominator is the one new piece; the numerator just repeats Halley. Carry the computation higher — by hand it turns ugly, so we ran it in a computer-algebra system through order 5 — and a clean pattern shows up. Define a sequence of polynomials in the scale ratios,

$$\begin{aligned}P_0 = P_1 &= 1,\\ P_2 &= 1 - \tfrac{1}{2}\rho_1,\\ P_3 &= 1 - \rho_1 + \tfrac{1}{6}\rho_1^2\rho_2.\end{aligned} \tag{14}$$

In terms of these, the order-d step is just

$$x_{t+1} = x_t - L_f\,\frac{P_{d-1}}{P_d}. \tag{15}$$

The numerator at each order is the denominator of the order before. So the family is one continued fraction in the scale ratios, read one rung deeper at each order — not a list of separate tricks. Each order folds in exactly one new ratio, ρ_{d-1}, riding the first scale that needs the d-th derivative.[2]

One case makes the structure vivid. On a pure exponential every rung equals L, every $\rho_k = 1$, and $P_d = 1/d!$, so (15) gives a step of $-L_f\,(1/(d-1)!)\big/(1/d!) = -d\,L_f$. Each order buys one more e-folding of progress on the scale-invariant function. That raises the obvious question: turn the order up without limit, and where does the step land?

A root is a pole, and the order is a dial you can turn

To answer it, look at what the step actually computes. Write the Taylor coefficients of f and of $1/f$ about x as $f_n = f^{(n)}(x)/n!$ and $g_n = (1/f)^{(n)}(x)/n!$. Then the order-d step (10) is exactly the ratio of two consecutive coefficients of $1/f$,

$$\text{step}_d = d\,\frac{(1/f)^{(d-1)}}{(1/f)^{(d)}} = \frac{g_{d-1}}{g_d}. \tag{16}$$

And those coefficients come free from one identity, $f \cdot (1/f) = 1$, read off order by order:

$$\sum_{k=0}^{n} f_k\, g_{n-k} = \delta_{n,0}. \tag{17}$$

This solves from the top down for each g_n in terms of the coefficients already in hand — the recursion the algorithm in Section 7 will use.

Now the picture. A root of f is a pole of $1/f$.

> *The Householder step is the ratio test ranging to that pole.*

Here is why, made concrete. Near a simple root $x^\star$, the reciprocal $1/f$ has a simple pole, and its coefficients about x grow like $g_n \propto (x^\star - x)^{-(n+1)}$. Their ratio is the distance itself, $g_{d-1}/g_d \to x^\star - x$ — the ratio test for the radius of convergence, which is the distance to the nearest singularity. With more than

[2] The whole table closes to one formula, $P_d = (-L_f)^d\, f\,(1/f)^{(d)}/d!$, which is just "take one more derivative of $1/f$" said in the scale variables. Its coefficients are a sum over the partitions of d by Faà di Bruno's formula [6]; we verified the telescoping through order 7.

one root, the step is a moment-ratio estimate of the nearest, sharpening as the order climbs onto higher coefficients. Table 1 shows it homing in for $f = (x-3)(x-7)$ started at $x = 0$.

Table 1: The step g_{d-1}/g_d for $f = (x-3)(x-7)$ started at $x = 0$, homing onto the nearest root $x^\star = 3$. Newton ($d = 1$) undershoots because the far root at 7 still pulls; each higher order leans on higher coefficients of $1/f$ and ranges the near root more sharply.

order d	1	2	3	4	5	6	7
g_{d-1}/g_d	2.100	2.658	2.860	2.941	2.975	2.989	2.995

That convergence has a closed-form limit, and finding it is where the continuous order comes in. The only discrete thing in the whole construction is the factorial $d!$. And the factorial is already an integral,

$$d! = \Gamma(d+1) = \int_0^\infty u^d \, e^{-u} \, \mathrm{d}u, \tag{18}$$

the d-th moment of a unit-rate exponential. Replace $d!$ by $\Gamma(d+1)$ and the order becomes a knob you can turn to any real value, even to infinity.

Turn it. Near the root, write $\lambda = x^\star - x$ for the distance. The coefficients $g_n = \lambda^{-(n+1)}$ are the normalized moments of an exponential density of rate λ, and summing them rebuilds $1/f$ as that density's Laplace transform [8],

$$\sum_{n \ge 0} g_n \, s^n = \int_0^\infty e^{-\lambda u} \, e^{su} \, \mathrm{d}u = \frac{1}{\lambda - s} = \frac{1}{f(x+s)}. \tag{19}$$

Push the order to any real ν and the gamma cancels, leaving the step equal to the distance at every order:

$$g_\nu = \frac{1}{\Gamma(\nu+1)} \int_0^\infty u^\nu \, e^{-\lambda u} \, \mathrm{d}u = \lambda^{-(\nu+1)}, \qquad \frac{g_{\nu-1}}{g_\nu} = \lambda = x^\star - x. \tag{20}$$

On a pure pole the method is exact the moment you write it down.

A general f is not a single pole. But for analytic f the reciprocal $1/f$ is meromorphic — a pole at every root — and its partial-fraction expansion writes it as the Laplace transform of one exponential per root, each decaying at the rate of its own distance $\lambda_j = x_j^\star - x$ and weighted by the residue w_j of $1/f$ there:

$$\frac{1}{f(x+s)} = \sum_j \frac{w_j}{\lambda_j - s} = \int_0^\infty e^{su} \sum_j w_j \, e^{-\lambda_j u} \, \mathrm{d}u, \qquad g_n = \sum_j w_j \, \lambda_j^{-(n+1)}. \tag{21}$$

Each coefficient is a residue-weighted blend of modes, and the step leans on the slowest — the nearest root — ever harder as the order climbs.[3] That is why the integer steps in the Table climb toward λ: $2.100 \to 2.995$ onto the root at $x^\star = 3$, now with a limit you can name.

This closes the circle. The exponential was the one function with a single fixed scale and no drift — every $\rho_k = 1$, the fixed point of Section 2. It returns here as the *law* whose Laplace transform is $1/f$ near the

[3]This needs analyticity, not mere smoothness: a C^∞ but non-analytic f has no convergent local series. The spectral weights are in general signed, even complex, not a probability density, and the one-sided transform is cleanest for roots ahead of x. For $f = (x-3)(x-7)$ about $x = 0$ the two modes are $\frac{1}{4}e^{-3u}$ and $-\frac{1}{4}e^{-7u}$, and the near one wins (Table 1). A root of multiplicity m contributes an Erlang mode $u^{m-1}e^{-\lambda u}$; a complex-conjugate pair, a damped oscillation.

root: the same function at both ends, once as the shape Newton steps along, once as the distribution the step reads.

An algorithm: step by the ratio test

Everything above turns into a short loop. The order-d Householder method never needs the symbolic d-th derivative of $1/f$; it needs only the coefficients g_n, and those come from the recursion (17) in $O(d^2)$ work. The step is one ratio.

```
input x, order d, tolerance ε, cap N
repeat
    f_k ← f^(k)(x)/k!     for k = 0,...,d
    g_0 ← 1/f_0
    g_n ← − f_0^{-1} Σ_{k=1}^{n} f_k g_{n−k}     for n = 1,...,d
    Δ ← g_{d−1} / g_d
    x ← x + Δ
until |Δ| < ε  or  iterations > N
return x
```

Three notes make it practical. *Firstly*, the cost per iteration is the d derivatives of f plus an $O(d^2)$ convolution — cheap, and the same derivatives any high-order method would want. *Secondly*, the order d is yours to set, and it buys convergence speed:

Proposition 1 (Convergence order). *If $x^\star$ is a simple root of a sufficiently smooth f, the order-d iteration (10) converges locally with order $d+1$: Newton quadratically, Halley cubically, order 3 quartically.*

We don't reprove this classical fact; see Householder [1] and Traub [2]. The structural reason is the one the paper has been making: the order-d step uses the scales $L_0, \ldots, L_{d-1}$ — derivatives through $f^{(d)}$ — and cancels the error in the distance-to-root expansion through that order. One more scale, one more ratio, one more order of convergence; and, on the exponential, one more e-folding.

Thirdly, the coefficients carry their own diagnostic. The ratios $g_0/g_1,\ g_1/g_2,\ \ldots,\ g_{d-1}/g_d$ — the row of Table 1 — should settle onto a single value as n grows, and that value is the distance to the nearest root. How fast they settle tells you whether the order is high enough to trust the step; when they have not settled, the nearest root has competition, and raising d sharpens the estimate.

Build the coefficients of $1/f$, take their last ratio, step. That is the whole method.

Conclusion

Read f/f' as a length and the classical root finders line up as one idea. Newton steps one characteristic scale toward the root. The scale drifts at rate $1-\rho_1$, and Halley divides by $1-\frac{1}{2}\rho_1$ to correct for it.

The order-d Householder method is the same leading scale L_f rescaled by a fixed rational function of the consecutive ratios $\rho_1, \ldots, \rho_{d-1}$ — one continued fraction, one rung deeper per order. That single step wears three faces: a clock reading the time-to-contact of a particle closing on a target; on a pure exponential the clean $-d\,L_f$, one e-folding per order; and, because a root of f is a pole of $1/f$, the ratio test ranging to that pole. Turn that last reading into a loop — build the coefficients of $1/f$, take their last ratio, step — and you have a root finder whose order is a dial. Higher-order root finding is reading the hierarchy of a function's own length scales one rung deeper.

Further Reading and References

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